

Dietary Glycemic Index, Glycemic Load, Sugar, and Fiber Intake in Association With Breast Cancer Risk: An Updated Meta-analysis

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Context: Several prospective cohort studies have investigated the association between glycemic index (GI), glycemic load (GL), dietary sugar, and total dietary fiber intake, with female breast cancer (BC) risk and reported inconsistent results. In the last decade, several large epidemiological studies have investigated these associations, suggesting the need to revisit the current body of evidence.

Objective: The aim of this study was to update a systematic review and meta-analysis conducted by Schlesinger et al in 2017 using recent scientific evidence published since 2015.

Data Sources: Publications indexed in PubMed, Embase, and The Cochrane Library were retrieved from the inception of the database up to January 2024.

Data Extraction: Two reviewers independently extracted data and assessed each study's quality.

Data Analysis: A random-effects model was used to estimate summary risk ratios (RRs) and 95% CIs for a meta-analysis that included 33 publications, with 26 prospective cohort studies cumulatively enrolling 2 212 645 women, among whom 79 777 were diagnosed with incident BC.

Results: Dietary GI and GL (highest vs lowest exposure intake) were both associated with 5% higher BC risk—RR (95% CI): 1.05 (1.01–1.09; $P = .007$) and 1.05 (0.97–1.13; $P = .24$), respectively. No clear associations were observed for sugar and total dietary fiber intake (highest vs lowest exposure intake)—RR (95% CI): 1.12 (0.95–1.11; $P = .16$) and 0.93 (0.86–1.00; $P = .05$), respectively. For the latter, the association was more pronounced among premenopausal women (RR: 0.78; 95% CI: 0.68–0.90; $P = .0008$).

Conclusion: This meta-analysis supports a significant positive association between high dietary GI intake and higher risk of BC and a significant inverse association between high dietary fiber intake and lower risk of BC. Interventions promoting a high-fiber and low-sugar diet may be useful components of BC-prevention strategies.

Systematic Review Registration: PROSPERO registration no. CRD42023463143.

Key words: carbohydrates, glycemic index, glycemic load, dietary sugar, dietary fiber, breast cancer, meta-analysis.

INTRODUCTION

Breast cancer (BC) incidence and mortality have been increasing worldwide over the last 3 decades, despite more favorable trends in many developed countries.¹ Nonmodifiable BC risk factors include older age, family history of BC, germline genetic polymorphisms, race/ethnicity, and menopausal status.² Modifiable BC risk factors include hormone replacement therapy, sedentary lifestyle, obesity, smoking, alcohol intake, and ultra-processed food consumption,¹ which remain primary targets for public health interventions involving behavioral and lifestyle modifications.³

The World Cancer Research Fund and American Institute for Cancer Research (WCRF/AICR) have designated certain aspects of diet as either protective or harmful for cancer risk.⁴ Specifically, dietary carbohydrate intake is recognized as a modifiable risk factor for cancer.⁵ Carbohydrates represent a broad macronutrient group that can be assessed using the glycemic index (GI).⁶ The GI is a scale of the blood glucose (ie, glycemic) response induced by consuming a particular food, expressed as a range from 0 to 100, in comparison to the response triggered by a reference food, such as white bread.⁷ Foods with a high GI have a higher composition of simple carbohydrates, such as added sugars, that are digested more rapidly, leading to a fast blood glucose spike, compared with foods with a low GI that have a higher composition of complex carbohydrates, such as dietary fiber.⁸ The glycemic load (GL) assesses the approximate effect at which a particular carbohydrate-containing food increases blood glucose levels, calculated by multiplying its GI with its carbohydrate content, expressed in grams (eg, g/100 g or g/serving).⁹ High GI and GL have been associated with a higher risk of cardiometabolic disease and certain cancers.^{10,11} However, no definitive conclusions have been reached regarding carbohydrate intake and BC risk. Foods with higher dietary fiber content, such as whole grains and non-starchy fruits/vegetables, may be associated with lower BC risk.¹² In contrast, ultra-processed foods containing higher levels of added sugars, such as sugar-sweetened beverages (SSBs), may be associated with higher BC risk.¹³

A 2008 meta-analysis of GI and GL in association with BC risk was largely inconclusive.¹⁴ A 2017 meta-analysis, conducted as part of the WCRF/AICR Continuous Update Project, assessed dose-dependent relationships and reported modest associations between GI and GL with BC risk. The latter meta-analysis included studies published through 2015.¹⁵ Several relevant studies have been published since then. With additional evidence available through January 2024, the aim is to refine estimates of the association between GI and

GL with BC risk and distinguish associations for carbohydrate-related dietary factors, such as intake of dietary sugar and fiber with BC risk.

METHODS

Protocol and Registration

This systematic review and meta-analysis follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Table S1)¹⁶ and the guidelines of the *Cochrane Handbook for Systematic Reviews of Interventions*, Version 6.4.¹⁷ The review protocol was registered as CRD42023463143 in the Prospective Register of Systematic Reviews (PROSPERO).¹⁸

Data Sources and Search Strategy

As in previous meta-analyses,^{14,15} this analysis followed the PICOS framework, where the population (P) included studies that enrolled female participants without BC history at baseline; exposures/intervention (E/I) reported the dietary intake of sugar (irrespective of source), total dietary fiber (irrespective of solubility), and derived GI and GL measures; comparison (C) involved evaluation of the risk ratio (RR) of BC for participants with high- vs low-exposure intake (eg, quantiles); outcome (O) was defined as clinically and/or histologically confirmed BC, irrespective of subtype; and study design (S) was limited to prospective cohort studies (Table 1).¹⁹ As part of a systematic literature search, 3 databases were interrogated: PubMed, Embase, and Cochrane Library.²⁰ Articles published from repository inception to January 15, 2024, were eligible for inclusion. The search query was composed of the following free-text and MeSH (Medical Subject Heading) terms: “glycemic index,” “glycemic load,” “dietary sugar” or “sugar intake” or “sugar-sweetened beverages” or “fiber,” and “breast neoplasms” or “breast tumors” or “breast cancer.” Details of the search strategy are summarized in Table S2. A manual search of references provided in articles identified in the primary search was also performed. Ethical approval and individual consent were obtained by investigators of the original studies.

Eligibility Criteria

Studies included in the meta-analysis were as follows: (1) prospective human cohort studies, (2) studies comparing the first incidence of female BC during follow-up and reporting adjusted estimates of RRs or hazard ratios (HRs) according to exposure quantiles with corresponding 95% CIs, (3) studies reporting clinically and/or

Table 1. PICOS Criteria for Inclusion of Studies

Parameter	Criterion
Population (P)	Studies that enrolled adult female participants (>18 y old) without BC history at baseline
Exposures/intervention (E/I)	Studies that reported exposures: (1) dietary intake of sugar, (2) dietary intake of total fiber, (3) GI, (4) GL
Comparison (C)	Involved evaluation of the risk ratio (RR) or hazard ratio (HR) of BC for participants with high- vs low-exposure intake (eg, quantiles)
Outcome (O)	Clinically and/or histologically confirmed incident BC, irrespective of subtype or disease classification
Study design (S)	Limited to prospective cohort studies for meta-analysis

Abbreviations: BC, breast cancer; GI, glycemic index; GL, glycemic load.

histologically confirmed incident BC cases, and (4) studies that published findings in English or Spanish language with full-text availability. When multiple publications reported on the same cohort study, the study with the largest number of BC events and overall cohort size was prioritized, even if it was not the most recent publication. Studies excluded from the meta-analysis were those that (1) only included preclinical models or functional analysis in silico or in vitro; (2) followed institutionalized or survivors or “diseased” individuals; and (3) review articles, abstracts, editorials, commentaries, letters to the editor, case reports, or study designs other than prospective cohort studies.

Study Selection and Data Extraction

Two authors (H.P.-M. and S.M.S.) separately identified, screened, and reviewed articles for eligibility. Data were uniformly extracted using a custom template (Table S3). Discrepancies were resolved by consensus with input from a third reviewer (M.N.P.). Abstracted data included the author(s) names, year of publication, country or region from which participants were enrolled, exposure definition(s), number at risk of BC at baseline, number of incident BC events, mean or median participant age at baseline, mean or median duration of follow-up, methods for exposure measurement and outcome ascertainment, covariates included in statistical analyses to control for confounding, RR or HR estimates, and 95% CIs (Table S3). Study quality and risk-of-bias assessment was performed by the lead author (H.P.-M.) (Table S4) using the Newcastle-Ottawa scale (NOS).²¹ If available, findings that reported subgroups (ie, menopausal, hormonal receptor status) were included in stratified meta-analysis.

Data Analysis

Extracted RR or HR estimates from maximally adjusted models were log-transformed, and standard errors were calculated. Primary analyses calculated a pooled measure of association from a random-effects model using a restricted maximum likelihood (REML) estimator.^{22,23}

The random-effects model was preferred to the fixed-effects model because it accounts for additional heterogeneity that may arise from differences between studies in the underlying population characteristics, follow-up duration, exposure measurement, and outcome definitions. However, fixed-effects models in secondary analyses were also considered to enhance comparability with estimates from previous meta-analyses.

Given that studies could report associations for different exposure quantiles (eg, tertiles, quintiles), a dose-response meta-analysis was used to establish a common scale for the association between exposure dose and disease risk following recommendations by Greenland and Longnecker²⁴ and Orsini et al.²⁵ Specifically, category-specific means or medians or midpoints (if a range was provided) of intake were used to define a grouped continuous variable, which was included in a linear model in primary analyses (Table S6). A cubic spline model with knots at the 5th, 50th, and 95th percentiles was used to test for nonlinearity. For dietary fiber studies, fiber intake was considered as “total fiber” if not specified otherwise. Heterogeneity between- and within-study estimates were explored through subgroup analysis and quantified using Cochrane’s Q test and the I^2 statistic.²⁶ Publication bias was assessed using funnel plots, Begg’s test, and Egger’s test.²⁷ A cumulative meta-analysis was performed by sequentially meta-analyzing estimates from studies in chronological order of publication date. A P value < .05 was considered statistically significant. All analyses were performed using R software (version 4.2.1) with *dosresmeta* (version 2.0.1), *meta* (version 7.0), and *metaphor* (version 4.4.0) packages.

RESULTS

Study Characteristics

In total, 3497 articles were identified from the literature search (Figure 1). After removing duplicates and screening titles and abstracts, 41 articles were identified for full-text evaluation. Eight articles were excluded upon full-text review (4 that were not prospective cohort

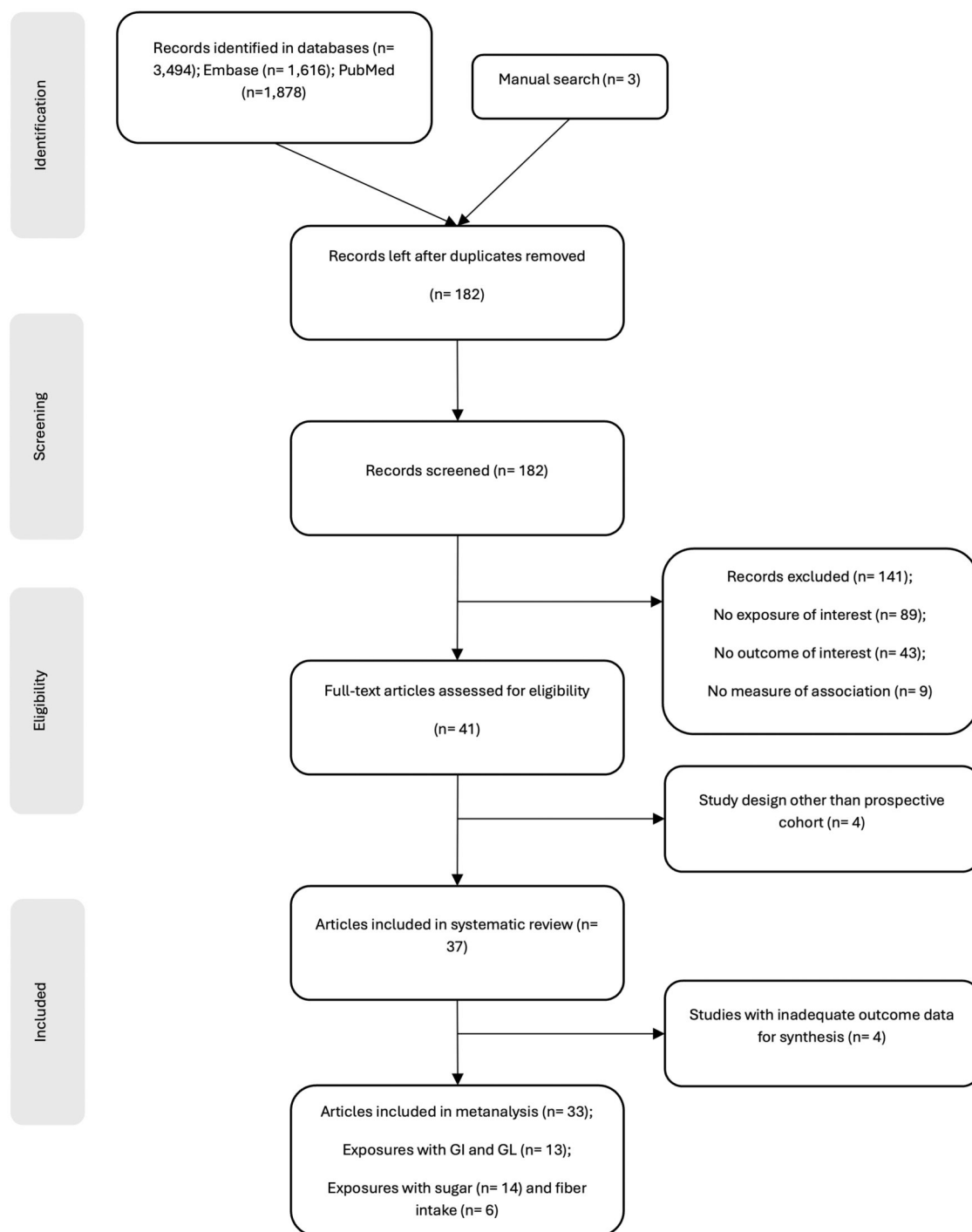


Figure 1. Flow Diagram of the Literature Search Process. Abbreviations: GI, glycemic index; GL, glycemic load

studies and 4 with inadequate outcome data for synthesis). The meta-analysis included 33 articles that reported associations between fiber intake, sugar intake, GI, GL, and BC risk (Table 2). These 33 articles described results from 26 distinct prospective cohorts that cumulatively enrolled 2 212 645 women between 18 and 79 years of age and 79 777 incident BC cases. The median of the reported average duration of the follow-up was

9.1 years. The median study quality score on the NOS was 5 (“fair” quality).

Association Between GI, GL, and BC Risk

Thirteen studies evaluated the association between GI and BC risk.^{28–42} Overall, higher GI (highest vs lowest exposure intake) was associated with higher BC risk (RR:

Table 2. Characteristics of Studies Included in the Meta-analysis of Cohort Studies Between Total Fiber Intake, Sugar Intake, Glycemic Index, and Glycemic Load With Breast Cancer Risk

Study (year)	Country	Study name	Dietary exposure	Units	Dietary intake tool	No. at risk	No. of breast cancer events	Age, mean or median, y	Follow-up, mean or median, y
Giles et al (2006) ⁵³	Australia	MCCS	Total sugars	1 serving/wk	FFQ: 121 items	12 273	324	—	9.1
Tasevska et al (2011) ⁶³	USA	National Institutes of Health (NIH)–AARP Diet and Health Study	Total sugars	g/1000 kcal	FFQ: 124 items	179 990	4793	—	7.2
Debras et al (2020) ⁵⁷	France	NutriNet-Santé Study	Total sugars	g/d	Validated web-based 24-h dietary records	101 279	2503	42.2	5.9
Kim et al (2017) ⁵⁸	South Korea	—	Sugary foods	—	FFQ: 16 food groups	5046	72	—	9.5
Fiolet et al (2018) ⁵⁵	France	NutriNet-Santé Study	Sugary products	g/d	Validated, web-based 24-h dietary records every 6 mo	82 159	739	42.8	5.4
Dunneram et al (2019) ⁶⁰	UK	UK Women's Cohort	Soft drinks	g/d	FFQ: 217 items	35 372	285	53.2	18
Chazelas et al (2019) ⁵⁶	France	NutriNet-Santé Study	Sugary drinks	100 mL/d	Repeated 24-h dietary records	101 257	2193	42.2	5.1
Hodge et al (2018) ⁵⁴	Australia	MCCS	Sugar-sweetened beverages (SSBs)	g/1000 kcal	FFQ: 121 items	36 539	946	—	11.6
Romanos-Nanclares et al (2019) ⁵⁹	Spain	SUN Project	Sugar-sweetened beverages (SSBs)	Servings/wk	FFQ: 136-item	10 713	100	33	10
Romanos-Nanclares et al (2021) ⁴⁵	USA	NHS I and II	Sugar-sweetened beverages (SSBs)	Servings/d	FFQ administered in NHS	175 798	11 379	47.8	26
Jonas et al (2003) ⁴⁴	USA	Cancer Prevention Study II Nutrition Cohort	Glycemic load	—	FFQ: 68 items	97 787	2884	—	5
Higginbotham et al (2004) ⁴³	USA	Women's Health Study	Glycemic index, glycemic load	Units/d	FFQ: 131 items	262 750	897	—	6.8
Nielsen et al (2005) ³⁴	Denmark	Diet, Cancer and Health cohort	Glycemic index	10 units/d	FFQ: 192 items	23 870	634	57	6.6
Silvera et al (2005) ³³	Canada	NBSS	Glycemic index	—	FFQ: 86 items	49 613	1450	54.8	16.6
Sieri et al (2007) ³⁶	Italy	ORDET study	Glycemic index	—	FFQ: 107 items	8926	207	—	11.5
Lajous et al (2008) ³⁰	France	E3N study cohort	Glycemic index	—	FFQ: 208 items	62 739	1812	53	9
Wen et al (2009) ³⁸	China	Shanghai Women's Health Study	Glycemic index	Units/d	FFQ	74 492	616	—	7.3
Larsson et al (2009) ³¹	Sweden	SMC	Glycemic index	—	FFQ: 67 and 96 items	61 433	2952	53.7	17.4
George et al (2009) ²⁸	USA	National Institutes of Health (NIH)–AARP Diet and Health Study	Glycemic index	—	FFQ: 124 items	183 535	5478	—	8
Romieu et al (2012) ³⁵	International	EPIC	Glycemic index	Units/d	Country-specific questionnaires	334 849	11 576	—	11.5
Sieri et al (2013) ³⁷	Italy	EPIC Italy	Glycemic index	—	Three FFQs: 1 for northern and central Italy (centers of Varese, Turin and	26 066	879	50.3	11

(continued)

Table 2. Continued

Study (year)	Country	Study name	Dietary exposure	Units	Dietary intake tool	No. at risk	No. of breast cancer events	Age, mean or median, y	Follow-up, mean or median, y
Farvid et al (2015) ⁴²	USA	NHS II	Glycemic index, glycemic load	Units/d	Florence), 1 for Ragusa, and 1 for Naples FFQ: 124 items	44 263	599	36.4	13
Makarem et al (2017) ³²	USA	FHS	Glycemic index	—	FFQ: 126 items	635	124	54.4	13.1
van Gils et al (2005) ⁸⁵	International	EPIC	Fruit juice	g/d	Country-specific questionnaires	285 526	3659	57	5.4
Shikany et al (2011) ⁴⁶	USA	WHI	Added sugars	g/d	FFQ designed for WHI	148 767	6115	63.6	8
Makarem et al (2018) ⁶⁴	USA	FHS	Fructose	% kcal	FFQ: 126 items	31 184	124	55.4	13.1
Key et al (2019) ⁶¹	UK	Million Women Study	Free sugar	% energy	FFQ: 130 items	691 571	29 005	59.9	12
Graham et al (1992) ⁵⁰	USA	New York State Cohort	Fiber intake	g	FFQ	3404	344	—	7
Suzuki et al (2008) ⁴⁹	Sweden	SMC	Fiber intake	g/d	FFQ: 87 and 97 items	51 823	2288	—	8.3
Maruti et al (2008) ⁵¹	USA	Vitamins and Lifestyle Cohort Study	Fiber intake	g/d	FFQ designed for WHI	28 586	507	61	5
Park et al (2009) ⁵²	USA	National Institutes of Health (NIH)—AARP Diet and Health Study	Fiber intake	g/d	FFQ: 124 items	185 598	5461	62	8
Narita et al (2017) ⁴⁸	Japan	Japan Public Health Center-based Prospective Study	Fiber intake	g/d	FFQ: 138 items	44 444	681	65	14
Heath et al (2020) ⁴⁷	International	EPIC	Fiber intake	1 SD	FFQ: 150 items	272 098	10 979	50	20.3

Abbreviations: EPIC, European Prospective Investigation into Cancer and Nutrition; FFQ, food-frequency questionnaire; FHS, Framingham Heart Study; MCCS, Melbourne Collaborative Cohort Study; NBSS, Canadian National Breast Screening Study; NHS, Nurses' Health Study; SMC, Swedish Mammography Cohort; SUN, Seguimiento Universidad de Navarra; WHI, Women's Health Initiative; E3N, Etude Epidémiologique auprès de femmes de la Mutuelle Générale de l'Education Nationale.

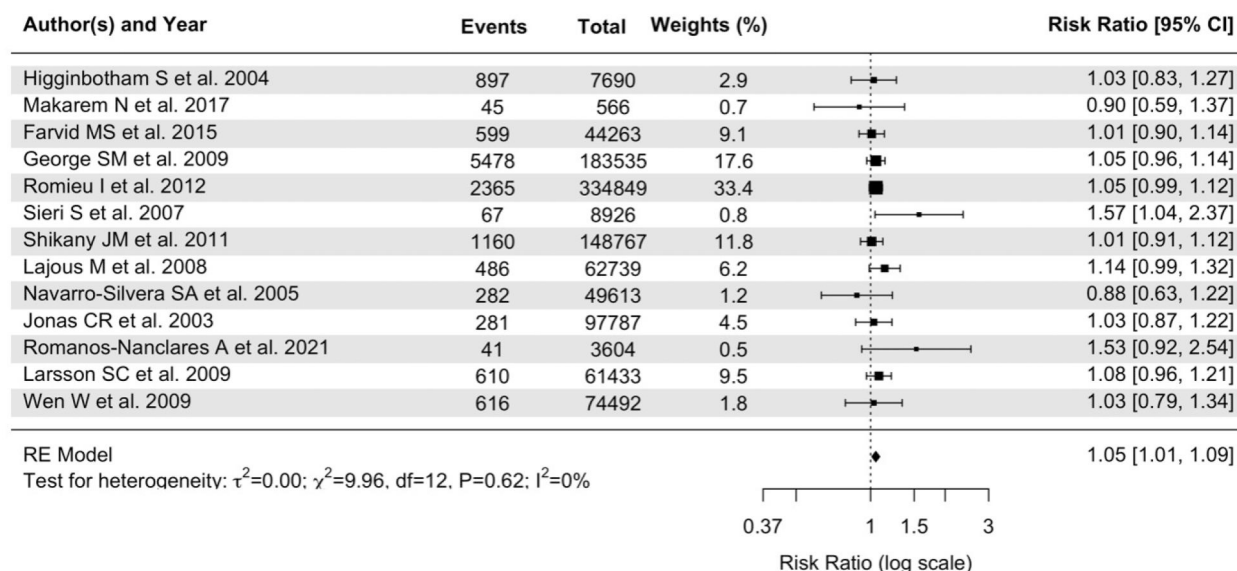


Figure 2. Forest Plot of Meta-analysis of Glycemic Index (GI) and Breast Cancer (BC) in Women. Individual studies are represented by risk ratio (RR) and 95% CI. Abbreviation: RE, Random Effects.

1.05; 95% CI: 1.01 to 1.09; $P=.007$, P_Q (Cochran's Q) = .6, $I^2 = 0.06\%$) (Figure 2). Seventeen studies evaluated the association between GI and BC risk.^{28–38,41–46} Overall, higher GI was associated with higher BC risk (RR: 1.05; 95% CI: 0.97 to 1.13; $P=.24$, $P_Q = .005$, $I^2 = 41\%$), although without achieving statistical significance and with modest heterogeneity.

Association Between Dietary Sugar and Fiber Intake and BC Risk

Eight studies evaluated total dietary fiber intake.^{47–52} Overall, total dietary fiber (highest vs lowest exposure intake) was associated with lower BC risk (RR: 0.93; 95% CI: 0.86 to 1.00; $P=.05$, $P_Q = .09$, $I^2 = 46.2\%$) (Figure 3). Twenty-one studies evaluated total, added, and/or free sugars from industrial or naturally occurring sources.^{45,46,53–70} When disaggregating the source of sugar exposure, 4 studies (5 publications)^{45,56,57,59,71} reported on SSB intake (RR: 1.12; 95% CI: 0.95 to 1.11; $P=.16$, $P_Q = .01$, $I^2 = 41\%$). Studies reporting exposure as total ($n=4$) or added sugars ($n=4$) were aggregated, showing no clear association with BC risk (RR: 1.06; 95% CI: 0.87 to 1.29; $P=.56$, $P_Q = .01$, $I^2 = 79.5\%$; and RR: 1.11; 95% CI: 0.90 to 1.37; $P=.32$, $P_Q = .03$, $I^2 = 82.3\%$, respectively).

Sensitivity and Stratified Analysis

Estimates from fixed-effects models were generally similar to those from random-effects models (Table S5). Leave-one-out analyses did not reveal excessive

influence of any single study. Some studies reported stratified results according to menopause status ($n=14$ premenopausal and $n=21$ postmenopausal). Among postmenopausal women, higher GI, GL, and SSB intake were associated with higher BC risk (RR: 1.09; 95% CI: 1.02 to 1.15; $P=.007$; $P_Q = .2$, $I^2 = 0.01\%$; RR: 1.06; 95% CI: 1.00 to 1.14; $P=.06$, $P_Q = .5$, $I^2 = 0.04\%$; and RR: 1.19; 95% CI: 1.00 to 1.43; $P=.05$, $P_Q = .1$, $I^2 = 45.8\%$, respectively). In premenopausal women, total dietary fiber intake was associated with lower BC risk (RR: 0.78; 95% CI: 0.68 to 0.90; $P=.0008$, $P_Q = .41$, $I^2 = 0.01\%$) compared with postmenopausal women (RR: 0.91; 95% CI: 0.83 to 0.98; $P=.018$, $P_Q = .45$, $I^2 = 0.01\%$) (Table S5). Dose-response meta-analysis (Table S6) identified an association between GI and BC risk in which each 1-unit increase in GI was associated with a 7% higher BC risk (RR: 1.07; 95% CI: 1.00 to 1.15; $P=.04$, $P_Q = .8$, $I^2 = 0.001\%$) (Figure S1). On the contrary, for total dietary fiber intake, each 10-g/day increase was associated with a 2% lower BC risk (RR: 0.88; 95% CI: 0.83 to 1.23; $P=.69$, $P_Q = .06$, $I^2 = 43.8\%$) (Figure S1); however, it was not significant. No evidence for nonlinearity was observed between exposures and outcomes.

Only a few studies stratified results according to tumor estrogen receptor (ER) and/or progesterone receptor (PR) status ($n=9$). For studies that included women with ER⁺ or PR⁺ status or both, no association was found to be statistically significant. Among those with ER[−] or PR[−] subtypes, GL had an RR of 1.29 (95% CI: 1.08 to 1.54; $P_Q = .47$, $I^2 = 5.1\%$). Three studies jointly stratified on menopausal and hormonal receptor

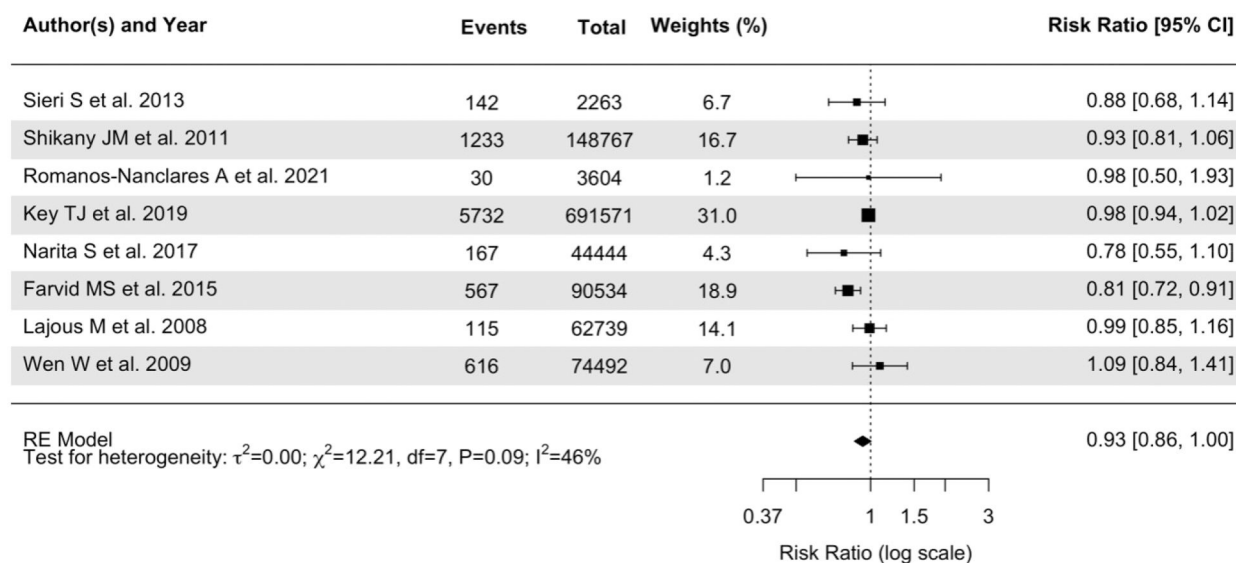


Figure 3. Forest Plot of Meta-analysis of Total Fiber Intake and Breast Cancer (BC) in Women. Individual studies are represented by risk ratio (RR) and 95% CI. Abbreviation: RE, Random Effects.

status, each reporting on ER/PR status among postmenopausal women, but none of the meta-analyzed associations for GI, GL, total dietary fiber, total sugar, and SSB intake were statistically significant for these subgroups. Across all exposures considered, there was no evidence of publication bias (Figures S3–S7, respectively).

Cumulative Meta-analysis

Cumulative meta-analysis revealed that statistically significant associations between GI and BC risk emerged in 2009 and persisted through 2021 with greater precision (Figure S8). More recently, a statistically significant association between total dietary fiber intake and BC risk appeared in 2017 and persisted through 2021 (Figure S10). No cumulative evidence of association was observed for GL (Figure S9), sugar intake (Figure S11), and SSB (Figure S12).

DISCUSSION

This updated meta-analysis of prospective cohort studies investigating the association of GI, GL, and carbohydrate dietary factors (ie, sugar or total dietary fiber intake) with BC risk showed a 5% higher risk for those with high-GI diets. Moreover, when stratified by menopause status, postmenopausal women had a higher RR than premenopausal women, suggesting that this group may be more susceptible to diet-associated BC risk. In contrast, higher fiber intake, irrespective of its solubility, tended to be associated with lower BC risk for both premenopausal and postmenopausal women. Although few studies reported associations for specific ER and PR

tumor subtypes, this meta-analysis revealed a positive association for GL that was stronger among women with ER[−] or PR[−] tumors specifically.

These results are comparable in direction and magnitude to the 2017 WCRF/AICR meta-analysis by Schlesinger et al.¹⁵ reporting that a 1-unit higher GI was associated with a 4% higher risk of BC overall and a 6% higher risk of BC among postmenopausal women specifically. Here, a beneficial association between total dietary fiber intake and BC risk was also observed, which was not reported by Schlesinger et al.¹⁵ Similarly, this dose–response meta-analysis showed a modest linear association between GI and BC risk, as reported previously.¹⁵ A recent meta-analysis of large cohorts evaluating the association between high-GI and high-GL food intake and type 2 diabetes and diabetes-related cancer incidence (including BC) showed a very similar risk estimate to the one reported here.⁷² Further evidence from cumulative meta-analysis, not considered as part of previously published meta-analyses, revealed that relevant studies published after 2015 had little impact on the overall magnitude and direction of these associations.

Higher total dietary fiber intake was associated with lower BC risk irrespective of menopausal or tumor receptor status. Results of previous studies investigating total dietary fiber intake have been inconclusive; some have reported an inverse association,⁷³ while others have found a null association.⁷⁴ A recent meta-analysis found that higher total dietary fiber intake (soluble or insoluble) was associated with an 8% lower BC risk for both pre- and postmenopausal women,¹² consistent with studies that have found inverse associations

between fruit intake and BC risk.⁷⁵ These findings are also comparable to those from the large meta-analysis of Reynolds et al,⁷⁶ suggesting that a higher intake of fiber and whole grains is associated with lower BC risk in a dose-response manner. Several mechanisms for the protective effect of fiber have been proposed, including modulation of ER activity in breast tissue by naturally occurring phytoestrogens (eg, flavonoids, lignans), fiber-induced carcinogen clearance,⁷⁷ or beneficial changes to microbiome composition.⁷⁸

Mechanistically, high-GI foods may promote carcinogenesis by sustaining an unfavorable elevated insulin response after eating.^{10,79} The potential proinflammatory effects associated with high added-sugar intake and increased risk of comorbidities, such as type 2 diabetes⁸⁰ and metabolic syndrome,⁸¹ may also increase cancer susceptibility.¹⁰ Diets rich in sucrose/fructose, such as those found in high-GI and ultra-processed foods, have been proposed to activate certain mechanistic pathways, including inflammation, glucose metabolism,⁷ immune suppression,⁸² and lipid metabolism.⁸³ In this meta-analysis, GL was not associated with BC risk, except for the ER⁻ tumor subtype. The inconsistency between findings for GI and GL and the high heterogeneity observed may suggest different effects for the type, quality, and quantity of carbohydrates affecting insulin release or varying hormonal response according to BC subtypes. Altogether, these findings suggest that dietary carbohydrates, specifically sugar and fiber intake, may contribute to BC risk, albeit with modest effect sizes.

This meta-analysis includes 12 new studies published after 2015 that were not incorporated into the previous meta-analysis by Schlesinger et al.¹⁵ This update results in a nearly doubled sample size (79 777 incident BCs among ~2.2 million women at risk, compared with 45 712 incident BCs among ~1.4 million at risk). Some recently published studies were substantially larger than previous studies. For example, the UK-based “Million Women Study”⁶¹ contributed nearly one-third of the total number of women at risk. Strict criteria were utilized to ensure included data were exclusively derived from prospective cohort studies. This provision helped limit potential bias from recall of prediagnostic dietary exposures for women diagnosed with BC, which can occur in case-control studies. The literature search strategy was comprehensive, acknowledging that articles not indexed in the major reference databases may have been missed. An ad hoc search for relevant gray literature (eg, using Google Scholar) did not identify obvious missed sources. Although GI and GL assessments capture some information on general dietary patterns, reporting food-component-specific risk estimates (ie, total dietary fiber and sugar) may be better for overall

interpretation, as these specify more directly the targets to seek or avoid when adopting a healthy diet. When designing diets, it is important to recognize variations in fiber quality across cereals, fruits, vegetables, and legumes; thus, low-GI dietary fibers including dried peas, beans, and lentils should be prioritized over those with medium- or higher GI values (eg, whole-wheat bread).

Despite observing modest heterogeneity in primary analyses, there are several limitations inherent in the individual studies that contributed to this meta-analysis. Only observational studies were included, and not all studies had the same degree of control for confounding variables. While most included studies adjusted for known BC risk factors, such as age at menarche, parity, oral contraceptive history, breastfeeding history, and family history of BC, relatively few adjusted for the intake of red meat or ultra-processed foods and the use of postmenopausal hormone replacement therapy. Thus, it is possible that, by aggregating differently adjusted estimates, the type I error may be increased. The adoption of healthy diets may be accompanied by favorable changes to physical activity levels and weight management. Few studies presented stratified results according to physical activity level (1 study),⁶⁷ weight change (5 studies),^{35,42,44,45,73} or body mass index (3 studies),^{29,41,42} thus limiting the ability to perform meta-analysis for these subgroups. It is possible that observations of protective effects of high-fiber or low-sugar diets are explained by the confounding effects of weight loss, uptake of other favorable lifestyle changes, or prevention of cardiometabolic comorbidities. The included studies often used different instruments for dietary assessment, and thus likely had differing degrees of measurement error. Most studies enrolled predominantly non-Hispanic White women, underscoring the need for new prospective cohort studies of these exposures among underrepresented minorities, particular because of the higher burden of BC in these populations.⁸⁴ Results by menopausal status and tumor receptor status have been reported here, and should be interpreted cautiously as these are based on a small number of studies and accordingly underpowered for certain subgroups.

CONCLUSION

This updated systematic review and meta-analysis provides evidence supporting the need for controlled-feeding studies or dietary-modification trial interventions to promote fiber-rich diets with low GI/GL. These human dietary intervention trials could help identify the optimal timing and duration of interventions and

collect biomarkers to understand the mechanistic pathways by which diet influences inflammatory, insulin, and glycemic responses. Dietary recommendations that promote fiber and low-GI/GL foods may be useful components of public health prevention strategies to decrease the risk of BC.

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Author Contributions. All authors contributed to the conception and design of the study. Material preparation, data collection, and data analyses were performed by H.P.-M. Data screening and study selection were conducted by H.P.-M. and S.M.S. H.P.-M., S.M.S., S.A.-M.D., M.R.K., and M.N.P. wrote the draft of the manuscript. All authors revised the manuscript and approved the submitted version.

Supplementary Material

[Supplementary Material](#) is available at *Nutrition Reviews* online.

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Conflicts of Interest

None declared.

Data Availability

All data used in this study are publicly available.

Code Availability

R scripts to reproduce study findings are available in Github/hpomares (https://github.com/hpomares/updated_MA_GI_GL_BC).

Ethics Approval

This meta-analysis used publicly available data; no ethical approval is required. Ethical approval and individual consent were obtained by investigators of the original studies.

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